



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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PROJECT TITLE

Adaptive AI Research Intelligence Platform for Automated Literature Review and Research Gap Discovery

ABSTRACT

The rapid growth of scientific publications across various domains has made it increasingly difficult for researchers and students to efficiently identify relevant literature, analyze research trends, and discover potential research gaps. Existing research tools primarily focus on keyword-based search and paper retrieval, requiring significant manual effort to review large volumes of literature. This creates challenges in conducting comprehensive literature reviews, understanding evolving research landscapes, and identifying emerging research opportunities.

The proposed **Adaptive AI Research Intelligence Platform for Automated Literature Review and Research Gap Discovery** addresses these challenges by providing an intelligent and adaptive system that automates literature analysis and knowledge extraction. The platform utilizes Artificial Intelligence (AI), Natural Language Processing (NLP), and Large Language Models (LLMs) to retrieve, process, classify, summarize, and analyze scientific research papers. It enables researchers to quickly understand research trends, identify relationships among studies, perform citation analysis, and discover unexplored areas within a domain. The system continuously adapts to user interests and feedback, providing personalized recommendations and updated insights as new research becomes available.

Research Gap: Current literature analysis systems mainly provide search and summarization capabilities but lack adaptive mechanisms for personalized research assistance, automated research gap identification, intelligent recommendation of future research directions, and continuous adaptation to changing research landscapes. Most existing solutions require significant manual intervention and do not effectively support data-driven research decision-making.

Software and Technologies Used: React.js, Spring Boot, PostgreSQL, REST APIs, Docker, Kubernetes, GitHub Actions, Argo CD, LangChain, ChromaDB Vector Database, and OpenAI-based Large Language Models.

Expected Outcome: The proposed system will enable researchers to perform faster and more effective literature reviews, automatically identify research gaps, generate concise summaries, discover emerging research trends, and receive adaptive recommendations for future research directions. The platform aims to reduce manual effort, improve research productivity, and support intelligent scientific knowledge discovery through adaptive software engineering principles.

This version is condensed to fit comfortably within a single abstract page while retaining the required sections: **Problem Statement, Research Gap, Software Used, and Expected Outcome.**

Signature of the Guide

HOD